

TECHNICAL & COMMERCIAL PROPOSAL

Integration and Reconfiguration of Chillers and
Chilled Water System

Qatar Distribution Company — Abu Hamour (QDC-AH)

Client / Facility	Qatar Distribution Company (QDC), Abu Hamour — managed by QRFM
Document Title	Chiller & CHW System Integration — Technical & Commercial Proposal
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Completion Period	4 weeks from award (night shift working, minimal disruption)
Validity	30 days from date of submission



1. Introduction

Metri Engineering Services W.L.L. (MES) is pleased to submit this technical and commercial proposal for the integration and reconfiguration of the chillers and chilled water (CHW) system at the Qatar Distribution Company retail and warehouse facility in Abu Hamour.

The facility is served by two independent CHW plants — System-1 (old) and System-2 (new) — which today operate in isolation, separated at the common header. QDC requires both plants to be integrated into a single coordinated system so that the newer plant provides capacity and redundancy to the ageing plant, improving reliability without disrupting operations.

MES proposes a **practical, measurement-led integration**: record the existing flows and pressures, physically interconnect the two headers, transfer circulation duty to the System-2 pumps, then re-measure and adjust the pump VFD to restore the required flows. The interconnection is carried out at night over two short shutdowns to keep cooling disruption to a minimum.

2. Existing System Summary

Parameter	SYSTEM-1 (Old)	SYSTEM-2 (New)
Chillers	Chiller-1 & Chiller-2	Chiller-3 & Chiller-4
Capacity	≈ 75 TR each (Trane, R-22)	≈ 150 TR each (Trane, R-134A)
Year	2001 (one compressor faulty)	2021
CHW Pumps	CHWP-1/2/3 — 11 kW (constant speed)	CHWP-4/5/6 — 18.5 kW (VFD)
LWT / EWT	6.2 / 8.5 °C (setpoint 6 °C)	7.8 / 10.8 °C (setpoint 7 °C)
CHW Pressure	50 / 80 PSI (in / out)	50 / 80 PSI (in / out)
Connected Load	≈ 110 TR	≈ 154 TR
Serves	Retail shop, offices, old warehouse	Extension warehouse

3. Scope of Work — Option 1 (Integration)

MES shall integrate all four chillers into a single operational system served by the System-2 pumps, through the following four activities:

3.1 Initial Flow & Pressure Measurement

- Measure the existing water flow of System-1 and System-2 at the pump DRV (double-regulating valve) test points.
- Record the existing pressure readings from the installed pressure gauges across both systems.
- These baseline readings define the target flows to be re-established after integration.

3.2 Pipe Modification & Interconnection

- Supply and installation of pipe and valves to physically connect the System-1 header to the System-2 header.
- The pipe connection (tie-in) is carried out **at night**, requiring a **2-hour shutdown of chillers/pumps over two consecutive nights** to minimize cooling disturbance to the facility.
- On completion, both systems' water circuits are opened. The System-1 pumps are isolated and switched off; the System-2 pumps run as 2 duty + 1 standby.
- The chillers from both systems remain running and are served by the System-2 pumps through the interconnected header, while the System-1 pumps stay isolated and off.

3.3 Re-Measurement & VFD Adjustment

- Re-measure the water flow at the DRVs and record the pressures under the integrated condition.
- Adjust the pump VFD frequency to match the required water flow against the initial baseline measurement, restoring balanced circulation across both systems.

3.4 Chilled Water Pipe Cladding

- Supply and installation of chilled water pipe cladding to the old (System-1) pipework to protect and finish the insulation.

Note: the System-2 pumps will circulate through both chiller sets via the interconnected header; the VFD adjustment in 3.3 ensures the combined flow and head are maintained. Setpoints of both chiller sets to be aligned to a common value for coordinated operation.

4. Option 2 — Decommissioning of System-1

As an alternative configuration, the System-1 chillers are permanently turned off and their isolation valves closed. The facility is then served entirely by the System-2 chillers and pumps through the interconnected header.

- Turn off System-1 chillers and close their isolation valves.
- Verify the water flow measurement and pressure readings to confirm the required duty is maintained across all loads by System-2 alone.

Option 2 uses the same physical pipe modification and cladding works as Option 1 and is offered **within the same commercial value**; the only difference is that the System-1 chillers are also isolated. This is the recommended long-term route given the age and R-22 obsolescence of the System-1 chillers.

5. Tools & Manpower

Instrument / Tool	Purpose
DRV commissioning manometer kit	Flow measurement at DRV test points
Pressure gauges	Pressure verification
Clamp meter	Pump motor current / VFD check
Welding set & pipe-working tools	Header tie-in and valve installation
Cladding & insulation tools	Pipe cladding works



6. Program (4 Weeks)

Activity	Week 1	Week 2	Weeks 3–4
Initial flow & pressure measurement	•		
Supply of pipe, valves & cladding materials	•	•	
Header tie-in — 2 consecutive nights (2-hr shutdowns)		•	
Open circuits, isolate System-1 pumps, run System-2 pumps (2 running +1 standby)		•	
Re-measurement & VFD adjustment			•
Chilled water pipe cladding			•
Final records & handover			•

Tie-in works are executed at night during pre-agreed windows to ensure minimal disruption to retail and warehouse operations.

7. Health, Safety & Quality

- Full PPE; daily toolbox talks; barricaded and signed work areas (“Work in Progress”).
- PTW with method statement and risk assessment prior to works; hot-work permit for welding/tie-in.
- Strict adherence to QDC site policies (no smoking / no alcohol), QRFM regulations and QCS standards.
- Calibrated instruments used for all flow and pressure measurements.

8. Assumptions & Exclusions

8.1 Assumptions

- Chillers, pumps and VFDs are in functional running condition; this scope is integration and cladding only.
- Existing DRVs, gauges and isolation valves are in good working condition and test points are existing.
- QDC provides safe access, power-isolation support and the agreed night-shutdown windows (2 hrs × 2 nights).
- Flow is maintained through the System-1 chiller evaporators via the interconnected header once the System-1 pumps are isolated.

8.2 Exclusions

- Repair or replacement of chiller compressors and any refrigerant works (R-22 / R-134A).
- Replacement of existing pumps, motors or VFDs unless found defective (priced separately).
- Major civil/structural works and permanent BMS integration.
- Authority / third-party / OEM certification fees unless specifically requested.



9. Commercial Proposal

The works are offered as a single lump-sum price for the integration of all four chillers into one operational system, including pipe modification, interconnection, re-balancing and chilled water pipe cladding.

9.1 Price breakdown

#	Scope item
1	Initial flow & pressure measurement (System-1 & System-2 at DRVs and gauges)
2	Pipe modification — supply & installation of pipe and valves to interconnect System-1 and System-2 headers, including night tie-in (2 × 2-hr shutdowns), opening of circuits and changeover to System-2 pumps (2 duty + 1 standby)
3	Re-measurement at DRVs, pressure recording and VFD frequency adjustment to match baseline flows
4	Chilled water pipe cladding for the old (System-1) pipework

TOTAL LUMP-SUM PRICE

QAR 94,000

9.2 Bill of Quantity

N°	Item Description	UoM	Qty	Total Price (QAR)
1	Budgetary cost for the integration of all four chillers to operate as one single system as per scope of work, including initial measurement, pipe modification and interconnection, re-measurement and VFD adjustment, and chilled water pipe cladding.	Each	1	94,000

9.3 Payment Terms

Milestone	% of Contract
Advance payment on award of Purchase Order	50%
Against completion of works	50%

- Validity: 30 days from submission.

MES is confident this practical approach delivers a stable, integrated chilled water system within the four-week program with minimal disruption to QDC operations. We remain available for a joint site visit and any technical clarification.

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